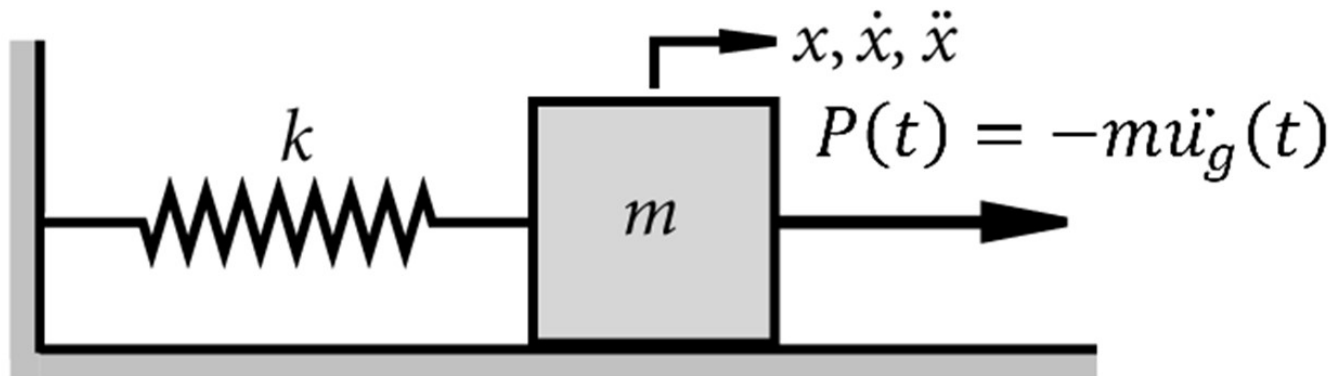


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SIMULATE THE SEISMIC RESPONSE OF A SINGLE- DEGREE-OF-FREEDOM (SDOF) STRUCTURE WITH COULOMB DRY FRICTION USING OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

Coulomb Dry Friction



$$m\ddot{x} + \mu F_n \operatorname{sgn}(\dot{x}) + kx = P(t) = -m\ddot{u}_g(t)$$



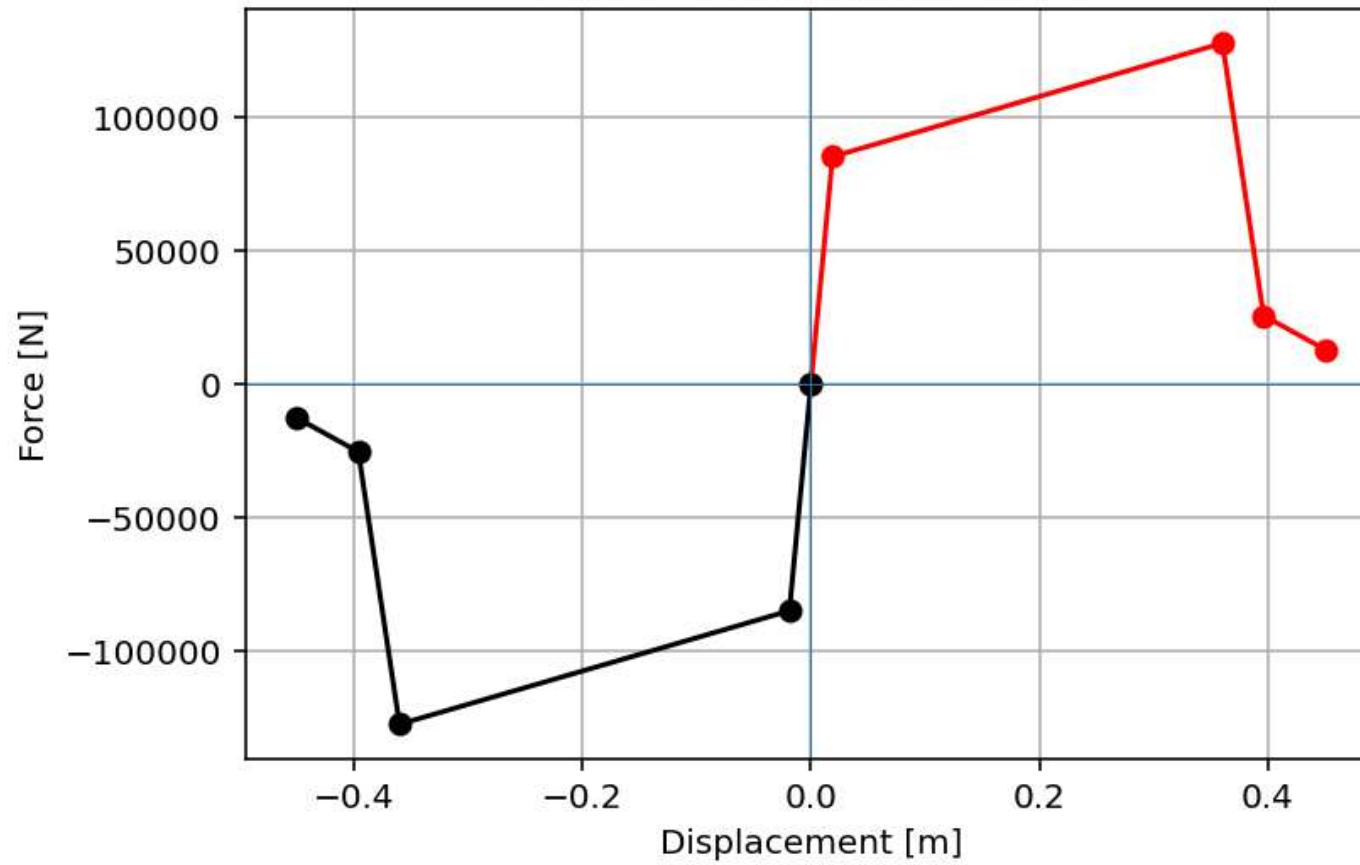
$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

Δ_u = Lateral Ultimate Displaement from Pushover Analysis

Force-Displacement Curve



Spdyer (Python 3.12)

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C:\Users\De\l\Desktop\OPENSEES_FILES\+SDOF_COULOMB...1\SDOF_SEISMIC_COULOMB_DRY_FRICTION_EXAMPLE_01.py

SDOF_SEISMIC_COULO...TION_EXAMPLE_01.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # SIMULATE THE SEISMIC RESPONSE OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) STRUCTURE W
4 # DRY FRICTION USING OPENSEES
5 #-----
6 # EXAMPLE 01
7 #-----
8 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
9 # EMAIL: salar.d.ghashghaei@gmail.com
10 #####
11 """
12 This Python code simulates a single-degree-of-freedom (SDOF) system with both a linear spr
13 The system's mass is 4000 tons, and the structural spring stiffness is 10000 kN/m. The Coul
14 A transient analysis is performed, subjecting the system to a sinusoidal acceleration input
15 The code records and plots the displacement, damper forces from both models, and hysteresis
16 """
17 # WIKIPEDIA
18 'https://en.wikipedia.org/wiki/Friction#:~:text=Coulomb%20friction%2C%20named%20after%20Cha
19 # BOOK: Differential Equations for Engineers - Wei-Chau Xie - CAMBRIDGE
20 'https://www.cambridge.org/core/books/differential-equations-for-engineers/1B8F1A62B6F98EB
21 # portwooddigital Website by Michael H. Scott
22 'https://portwooddigital.com/2024/02/10/two-sprung-masses-and-some-friction-force/'
23 #%%
24 import openseespy.opensees as ops
25 import numpy as np
26 import matplotlib.pyplot as plt
27 import time as TI
28 import ANALYSIS_FUNCTION as S01
29 import PERIOD_FUN as S02
30 import DAMPING_RATIO_FUN as S05
31 import EIGENVALUE_ANALYSIS_FUN as S06
32 #%%
33 # Define parameters (units: m, N)
34 FY = 85000.0 # [N] Yield Force of Structure
```

Period of Structure vs Displacement During Free Vibration Analysis

Legend:

- ELASTIC PERIOD - MIN VALUES: Min: 0.662 (s) - Mean: 0.662 (s) - Max: 0.662 (s)
- ELASTIC PERIOD - MAX VALUES: Min: 0.662 (s) - Mean: 0.662 (s) - Max: 0.662 (s)
- INELASTIC PERIOD - MIN VALUES: Min: 0.662 (s) - Mean: 0.729 (s) - Max: 3.980 (s)
- INELASTIC PERIOD - MAX VALUES: Min: 0.662 (s) - Mean: 0.729 (s) - Max: 3.980 (s)

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Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 14, Col 159 UTF-8 CRLF RW Mem 43%

Equivalent viscous damping ratio - Seismic Hysteresis

